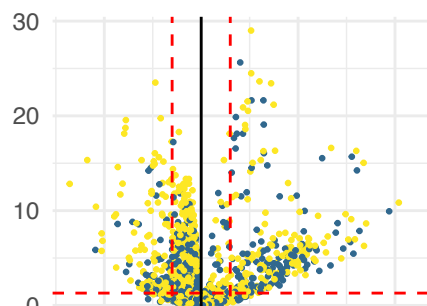


A

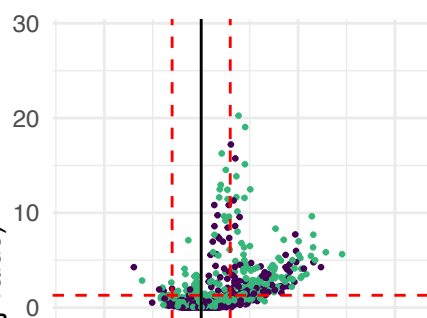
Protease	Positional Specificity	Amino Acid Specificity	Ref
Cathepsin A	Carboxypeptidase	Removes hydrophobic amino acids (e.g. Phe and Leu) from the C-terminus of substrates when additional hydrophobic amino acids are present in P1	25
Cathepsin B	Dipeptidyl carboxypeptidase	Removes dipeptides from the C-terminus when either positively charged (Arg, Lys) or hydrophobic amino acids are in the P1 and P2 position	26
Cathepsin C	Dipeptidyl aminopeptidase	Removes dipeptides from the N-terminus with broad specificity	8
Cathepsin D	Endopeptidase	Cleaves between hydrophobic amino acids (e.g. Phe, Leu, Tyr)	27

B

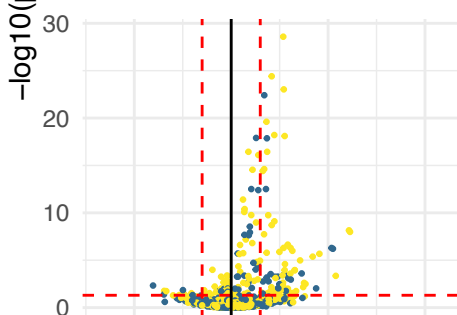
Cathepsin A



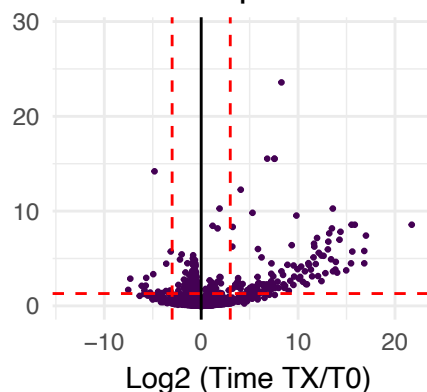
Cathepsin B



Cathepsin C



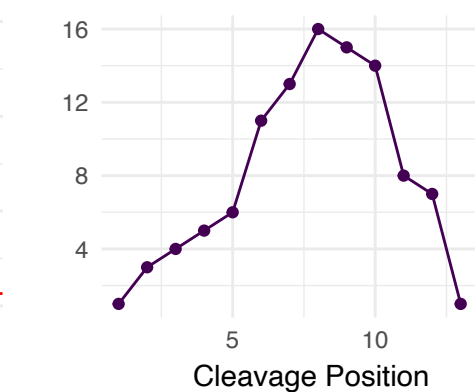
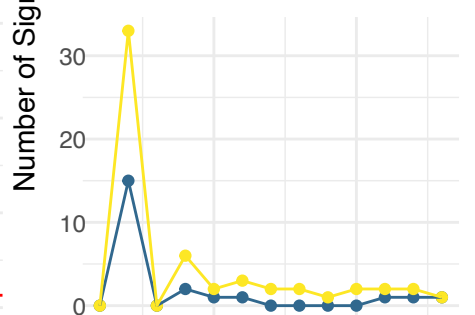
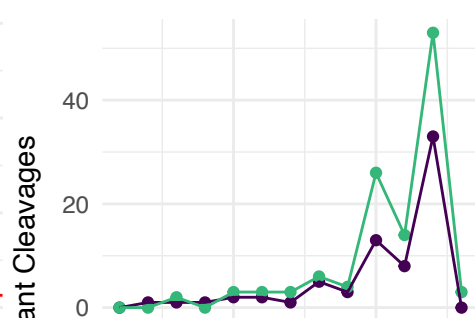
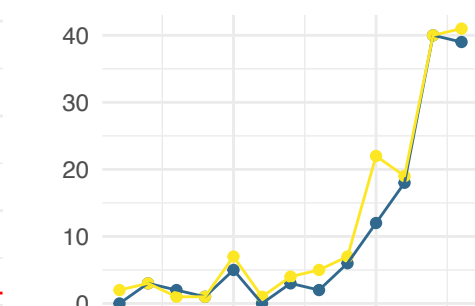
Cathepsin D



Time (minutes)

● 15 ● 30 ● 60 ● 90

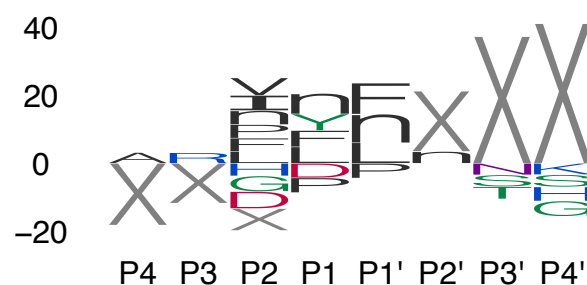
C



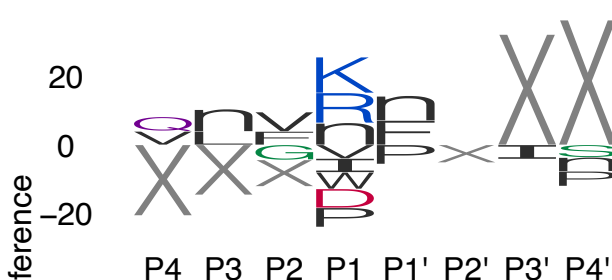
Cleavage Position

D

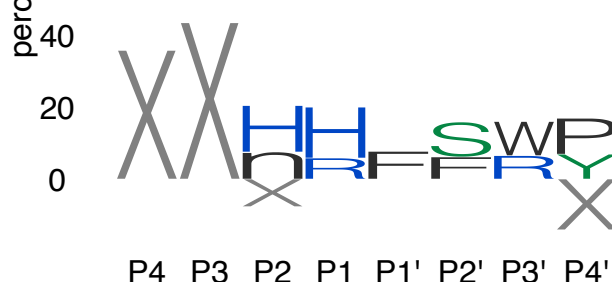
n = 165



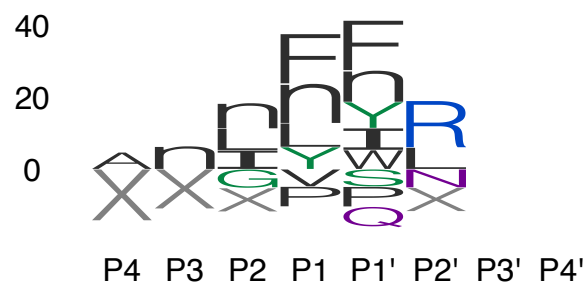
n = 118



n = 56



n = 92



Chemistry

■ Acidic ■ Hydrophobic ■ Past Terminus
 ■ Basic ■ Neutral ■ Polar